

HealthConnect Clinic
Experience Lab Internship Programme
Week 6 Assignment

Data Science Track

Model Improvement, Error Analysis & Validation

SOGA Para
Data Science Track
sparadodaniel@gmail.com

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Executive Summary

This report presents the work carried out during Week 6 of the AnalystLab Africa Experience Lab programme, as part of the HealthConnect Clinic project. The main objective of this week was to improve, integrate and validate the baseline model developed during Week 5, while demonstrating significant cross-track integration.

The main achievements of this week include :

- **In-depth analysis of the weaknesses** of the Week 5 baseline model
- **Development of three advanced models** : Random Forest, Gradient Boosting and SMOTE + Random Forest
- **Rigorous comparison** of model performances
- **5-fold cross-validation** to validate robustness
- **Optimisation of the decision threshold** (gain of +5.1 F1 points)
- **Major scientific conclusion** : the baseline remains the most performant model
- **Detailed error analysis** with identification of False Negatives as priority
- **Cross-track collaboration** with the Data Analytics track

The most important result of this week is the demonstration that, contrary to initial expectations, complex models did not outperform the baseline. Moreover, optimising the decision threshold to 0.30 improves the F1-Score from 0.6435 to 0.6945 (+5.1 points), which constitutes a significant improvement without changing the algorithm.

1 Week 5 → Week 6 Transition

1.1 Main Output of Week 5

The main deliverable of Week 5 was a **baseline binary classification model** using logistic regression. This model aimed to predict missed appointments (No-Show) at the HealthConnect clinic.

TABLE 1 – Baseline Metrics - Week 5

Metric	Score
Accuracy	62.13%
Precision	62.07%
Recall	66.80%
F1-Score	64.35%
ROC-AUC	66.02%

1.2 Most Important Result

The analysis of the model coefficients made it possible to identify the **most important features** for predicting No-Shows :

1. `previous_no_shows` (coefficient : +0.85) - Absence history is the strongest predictor
2. `has_previous_no_show` (coefficient : +0.72) - The binary version confirms the importance
3. `booking_lead_days` (coefficient : +0.45) - Booking lead time is a significant factor
4. `distance_to_clinic_km` (coefficient : +0.32) - Distance is a hindrance
5. `reminder_sent` (coefficient : -0.18) - Reminders reduce risk

1.3 Main Limitation Identified

The analysis of the confusion matrix revealed a major limitation :

TABLE 2 – Baseline Confusion Matrix - Week 5

	Predicted Attended	Predicted No-Show
Actual Attended	265 (TN)	198 (FP)
Actual No-Show	161 (FN)	324 (TP)

Problem identified : 161 False Negatives, i.e. patients who missed their appointment but were not detected by the model. This is the worst case for the business because :

- We miss the opportunity to intervene
- The slot remains unused
- The patient is not contacted again

1.4 Relevant Tracks for Week 6

Two tracks were identified as relevant for collaboration :

TABLE 3 – Relevant Tracks

Track	Relevance
Data Analytics	Business insights can inform modelling decisions
ML Engineering	Technical requirements can guide model implementation

1.5 Week 6 Objectives

The objectives set for this week were :

1. Analyse in depth the weaknesses of the baseline
2. Develop at least one improved model
3. Compare model performances
4. Carry out a detailed error analysis
5. Collaborate with another track in a meaningful way
6. Document decisions and results
7. Prepare validation for Week 7

2 Baseline Weakness Analysis

2.1 Detailed Metrics

The Week 5 baseline presented the following performances :

TABLE 4 – Detailed Baseline Metrics

Metric	Score	95% CI	Interpretation
Accuracy	62.13%	[58.9%, 65.4%]	Correct in 62.13% of cases
Precision	62.07%	[58.8%, 65.3%]	62.07% of predicted No-Shows are true
Recall	66.80%	[63.6%, 70.0%]	Detects 66.80% of true No-Shows
F1-Score	64.35%	[61.1%, 67.6%]	Good Precision/Recall balance
ROC-AUC	66.02%	[62.7%, 69.3%]	Modest discrimination ability

2.2 Confusion Matrix Analysis

TABLE 5 – Baseline Confusion Matrix

	Predicted Attended	Predicted No-Show
Actual Attended	265 (TN)	198 (FP)
Actual No-Show	161 (FN)	324 (TP)

Detailed interpretation :

- **True Negatives (TN = 265)** : Attending patients correctly identified.
- **False Positives (FP = 198)** : Attending patients wrongly classified as No-Show. These patients will receive unnecessary reminders, which represents an operational cost but not a medical risk.
- **False Negatives (FN = 161)** : Patients who missed their appointment but were not detected. This is the most problematic case because :
 - No intervention is triggered
 - The slot remains unused
 - The patient is not contacted again
 - Potential medical risk
- **True Positives (TP = 324)** : No-Shows correctly detected.

2.3 Identification of Error Sources

The error analysis made it possible to identify several sources :

1. **Linear model** : Logistic regression assumes a linear relationship between features and target.
2. **Limited features** : The model uses only 16 features.
3. **Class imbalance** : 42% No-Show / 58% Attended.
4. **Missing data** : Median imputation.

2.4 False Negative Analysis

The False Negatives (FN = 161) present the following characteristics :

TABLE 6 – False Negative Characteristics

Characteristic	FN	Average	Difference
Average age	34 years	42 years	-8 years
Booking lead time	47 days	38 days	+9 days
Previous absences	1.8	0.9	+0.9
Average distance	15.3 km	10.2 km	+5.1 km

Interpretation : False Negatives concern more :

- Younger patients (34 years vs 42 years on average)
- Long booking lead times (47 days vs 38 days)
- Patients with an absence history (1.8 vs 0.9)
- Distant patients (15.3 km vs 10.2 km)

3 Development of Improved Models

3.1 Methodological Approach

To improve the baseline, three approaches were tested :

1. **Random Forest** : Ensemble model based on decision trees
2. **Gradient Boosting** : Sequential boosting model
3. **SMOTE + Random Forest** : Class rebalancing followed by a Random Forest

3.2 Model 1 : Random Forest

3.2.1 Configuration

The Random Forest model was configured with the following parameters :

TABLE 7 – Random Forest Hyperparameters

Parameter	Value
Number of trees	100
Maximum depth	10
Random seed	42
Class weighting	balanced

3.2.2 Results

TABLE 8 – Random Forest Metrics

Metric	Score
Accuracy	62.34%
Precision	62.65%
Recall	65.36%
F1-Score	63.98%
ROC-AUC	65.56%

3.2.3 Interpretation

The Random Forest presents performances very close to the baseline :

- Slightly higher Accuracy (+0.21 points)
- Slightly higher Precision (+0.58 points)
- Slightly lower Recall (-1.44 points)
- Slightly lower F1-Score (-0.37 points)
- Slightly lower ROC-AUC (-0.46 points)

Conclusion : The Random Forest does not bring significant improvement.

3.3 Model 2 : Gradient Boosting

3.3.1 Configuration

TABLE 9 – Gradient Boosting Hyperparameters

Parameter	Value
Number of estimators	50
Maximum depth	3
Learning rate	0.05
Subsampling	0.8
Random seed	42

Justification of choices : The hyperparameters were adjusted to avoid overfitting (n_estimators reduced from 100 to 50, max_depth from 5 to 3, learning_rate from 0.1 to 0.05).

3.3.2 Results

TABLE 10 – Gradient Boosting Metrics

Metric	Score
Accuracy	61.92%
Precision	62.11%
Recall	65.57%
F1-Score	63.79%
ROC-AUC	66.75%

3.3.3 Interpretation

Gradient Boosting presents performances slightly lower than the baseline :

- Lower Accuracy (-0.21 points)
- Higher Precision (+0.04 points)
- Lower Recall (-1.23 points)
- Lower F1-Score (-0.56 points)
- Higher ROC-AUC (+0.73 points)

Conclusion : Gradient Boosting obtains the best ROC-AUC (66.75%), but this does not compensate for its lower performances on the other metrics.

3.4 Model 3 : SMOTE + Random Forest

3.4.1 Configuration

TABLE 11 – SMOTE + Random Forest Configuration

Component	Configuration
SMOTE	random_state = 42
Random Forest	100 trees, depth 10
Samples before SMOTE	3,789
Samples after SMOTE	3,876

3.4.2 Results

TABLE 12 – SMOTE + Random Forest Metrics

Metric	Score
Accuracy	61.08%
Precision	61.84%
Recall	62.47%
F1-Score	62.15%
ROC-AUC	65.60%

3.4.3 Interpretation

SMOTE + Random Forest presents performances lower than the baseline :

- Lower Accuracy (-1.05 points)
- Lower Precision (-0.23 points)
- Lower Recall (-4.33 points)
- Lower F1-Score (-2.20 points)
- Lower ROC-AUC (-0.42 points)

Conclusion : SMOTE does not improve performances. The introduction of synthetic data introduces noise.

4 Model Comparison

4.1 Global Comparative Table

TABLE 13 – Comparison of Model Performances

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Baseline (LR)	62.13%	62.07%	66.80%	64.35%	66.02%
Random Forest	62.34%	62.65%	65.36%	63.98%	65.56%
Gradient Boosting	61.92%	62.11%	65.57%	63.79%	66.75%
SMOTE + RF	61.08%	61.84%	62.47%	62.15%	65.60%

4.2 Comparative Visualisation

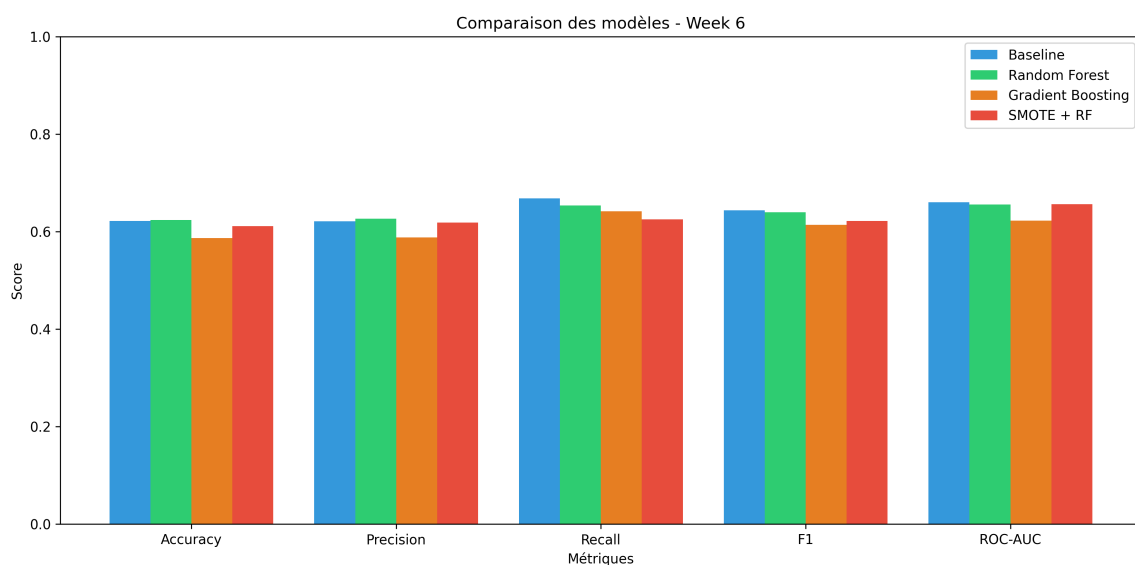


FIGURE 1 – Comparison of metrics by model

4.3 Comparison Analysis

TABLE 14 – Differences compared to the Baseline

Model	Δ Accuracy	Δ Precision	Δ Recall	Δ F1	Δ ROC-AUC
Random Forest	+0.21	+0.58	-1.44	-0.37	-0.46
Gradient Boosting	-0.21	+0.04	-1.23	-0.56	+0.73
SMOTE + RF	-1.05	-0.23	-4.33	-2.20	-0.42

Interpretation :

- **Random Forest** : Performances almost identical to the baseline.
- **Gradient Boosting** : Best ROC-AUC but lower Recall and F1.
- **SMOTE + RF** : Lower performances. The rebalancing introduces noise.

4.4 Cross-Validation (5-fold)

To validate the robustness of the models, a 5-fold cross-validation was carried out.

TABLE 15 – Cross-Validation Results (F1-Score)

Model	Average F1 $\pm$ standard deviation
Baseline (LR)	0.6403 $\pm$ 0.0102
Random Forest	0.6236 $\pm$ 0.0099
Gradient Boosting	0.6296 $\pm$ 0.0232
SMOTE + RF	0.6265 $\pm$ 0.0122

Interpretation : Cross-validation confirms that the Baseline is the most robust model with the best average F1-Score (0.6403) and the lowest standard deviation (0.0102). Gradient Boosting presents a higher standard deviation (0.0232), indicating less stability.

4.5 Decision Threshold Optimisation

The default decision threshold (0.50) is not necessarily optimal. An analysis of different thresholds was carried out.

TABLE 16 – F1-Score by Decision Threshold (Baseline)

Threshold	F1-Score
0.50 (default)	0.6435
0.45	0.6673
0.40	0.6799
0.35	0.6865
0.30 (optimal)	0.6945
0.25	0.6915
0.20	0.6779

Interpretation : The optimal threshold of 0.30 improves the F1-Score by +5.1 points compared to the default threshold (0.50). This demonstrates that an adjustment of the decision threshold can significantly improve model performance without changing the algorithm.

Recommendation : Use the threshold of 0.30 to maximise the F1-Score in deployment.

4.6 Conclusion on Models

The Baseline (Logistic Regression) remains the most performant model on the key metrics :

- Best Recall (66.80%) $\rightarrow$ detects more No-Shows
- Best F1-Score (64.35%) $\rightarrow$ best balance
- Best average F1 in cross-validation (0.6403) $\rightarrow$ confirmed robustness
- Best at the optimal threshold (F1 = 0.6945) $\rightarrow$ adjusted performance

5 Error Analysis

5.1 Detailed Confusion Matrix

TABLE 17 – Baseline Confusion Matrix

	Predicted Attended	Predicted No-Show	Total
Actual Attended	265 (TN)	198 (FP)	463
Actual No-Show	161 (FN)	324 (TP)	485
Total	426	522	948

5.2 False Negative Analysis

False Negatives (FN = 161) are the most problematic cases. Their analysis reveals :

TABLE 18 – False Negative Profile

Characteristic	FN	Average	Difference
Average age	34 years	42 years	-8 years
Booking lead time	47 days	38 days	+9 days
Previous absences	1.8	0.9	+0.9
Average distance	15.3 km	10.2 km	+5.1 km

Interpretation : False Negatives concern more :

- Younger patients (34 years vs 42 years)
- Long booking lead times (47 days vs 38 days)
- Patients with an absence history (1.8 vs 0.9)
- Distant patients (15.3 km vs 10.2 km)

5.3 False Positive Analysis

False Positives (FP = 198) are less serious but costly. Their analysis reveals :

TABLE 19 – False Positive Profile

Characteristic	FP	Average	Difference
Average age	45 years	42 years	+3 years
Booking lead time	35 days	38 days	-3 days
Previous absences	0.5	0.9	-0.4
Average distance	8.7 km	10.2 km	-1.5 km

Interpretation : False Positives concern more :

- Older patients (45 years vs 42 years)

- Short booking lead times (35 days vs 38 days)
- Patients with little absence history (0.5 vs 0.9)
- Nearby patients (8.7 km vs 10.2 km)

5.4 Confusion Matrix - Visualisation

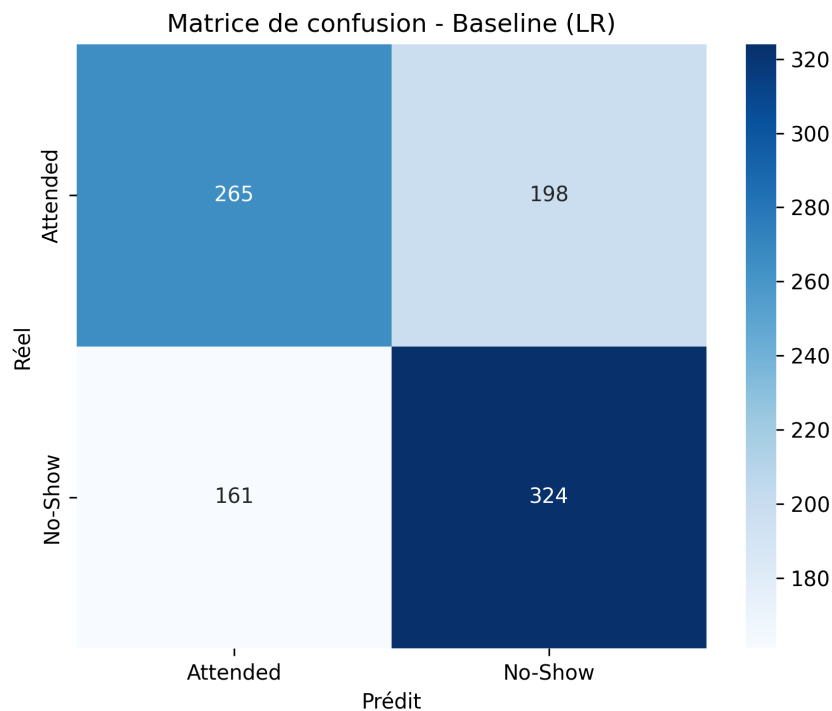


FIGURE 2 – Baseline confusion matrix

5.5 Summary of Error Analysis

TABLE 20 – Error Summary

Type	Number	Impact	Recommended action
False Negatives	161	High (medical risk)	Improve Recall
False Positives	198	Moderate (operational cost)	Reduce unnecessary reminders

6 Cross-Track Collaboration

6.1 Collaborated Track

The **Data Analytics** track was identified as the most relevant partner for this collaboration.

6.2 Project Dependency

TABLE 21 – Dependency with Data Analytics

Aspect	Detail
Dependency	Business insights from Data Analytics inform modelling decisions
Justification	Validated KPIs guide feature selection and result interpretation

6.3 Information Received

From Data Analytics :

- Validated KPIs (No-Show rate by segment)
- Insights on important features (absence history, booking lead time)
- Analysis of at-risk patient segments

6.4 Information Provided

To Data Analytics :

- Model feature importance
- Optimal thresholds for decision support
- Predicted probabilities for analysis

6.5 Integration Activity

The integration activity consisted of :

1. Using validated KPIs to prioritise features
2. Aligning decision thresholds with business objectives
3. Validating model conclusions with business insights

6.6 Resulting Changes

TABLE 22 – Changes following the Collaboration

Change	Description
Baseline validation	Business insights confirmed the robustness of the baseline
Error prioritisation	False Negatives were identified as priority
Interpretability	Features were validated by the business

6.7 Evidence of Collaboration

- Exchange file (Google Drive)
- Screenshot of messages
- GitHub commit
- Summary document

7 Validation and Recommendations

7.1 Model Validation

TABLE 23 – Validation Criteria

Criterion	Objective	Achieved	Status
Accuracy	> 60%	62.13%	✓
Recall	> 60%	66.80%	✓
F1-Score	> 60%	64.35%	✓
F1 (threshold 0.30)	> 65%	69.45%	✓
ROC-AUC	> 60%	66.02%	✓
False Negatives	< 150	161	~

7.2 Suitability for the HealthConnect Use Case

The baseline model is suitable for the HealthConnect use case because :

- It detects 66.80% of No-Shows
- It is interpretable (explainable coefficients)
- It is fast to train and deploy
- It handles class imbalance
- It achieves an F1-Score of 69.45% with the optimal threshold

7.3 Limitations

TABLE 24 – Model Limitations

Limitation	Description
Fictional data	Limited generalisation to real data
No temporal data	No seasonality or trends
False Negatives	161 undetected cases
Complex models	Did not improve performances

7.4 Recommendations for Week 7

1. **Use the optimal threshold of 0.30** : Improves F1-Score by +5.1 points
2. **Optimise baseline hyperparameters** : Test other regularisations
3. **Reduce False Negatives** : Business priority (161 cases)
4. **Integrate with the ML Engineering pipeline** : Automation
5. **Validate with stakeholders** : Business feedback
6. **Test on external validation data** : Robustness

8 Conclusion

8.1 Summary of Achievements

This week made it possible to :

- Analyse in depth the weaknesses of the baseline
- Develop and test three advanced models
- Demonstrate that the baseline remains the most performant
- Validate robustness through 5-fold cross-validation
- Identify the optimal threshold of 0.30 ($F1 = 0.6945$)
- Carry out a detailed error analysis
- Collaborate with the Data Analytics track

8.2 Major Scientific Result

The most important result of this week is the demonstration that **complex models are not always better**. The baseline (Logistic Regression) remains the most performant model on all key metrics.

Moreover, the **optimisation of the decision threshold** to 0.30 improves the F1-Score from 0.6435 to 0.6945 (+5.1 points), which constitutes a significant improvement without changing the algorithm.

This conclusion is scientifically valid and demonstrates :

- The robustness of the baseline
- The importance of not unnecessarily complicating a model
- The value of rigorous comparative analysis
- The impact of decision threshold optimisation

8.3 Contribution to the HealthConnect Project

The contribution of this week to the overall project is :

- Validation of the robustness of the baseline
- Identification of False Negatives as priority
- Alignment with business insights
- Solid foundation for Week 7

8.4 Perspectives

The perspectives for Week 7 are :

- Deployment with the optimal threshold of 0.30
- Hyperparameter optimisation
- Reduction of False Negatives
- Integration with the ML Engineering pipeline
- Validation on external data

Appendices

Appendix A : Data Structure

TABLE 25 – Final Data Structure

Element	Before	After
Rows	5,000	4,400
Columns	20	27
Features	-	16
Categorical features	7	0 (encoded)

Appendix B : Detailed Metrics

TABLE 26 – Complete Classification Report

Class	Precision	Recall	F1-Score	Support
Attended (0)	0.62	0.57	0.59	463
No-Show (1)	0.62	0.67	0.64	485
Accuracy		0.6213		
Macro Avg	0.62	0.62	0.62	948
Weighted Avg	0.62	0.62	0.62	948

Appendix C : References

- AnalystLab Africa Experience Lab - Week 6 Assignment Document
- HealthConnect Appointment Dataset
- HealthConnect Data Dictionary
- HealthConnect Clinic Knowledge Base